ECE 410 | Codefest 7 CMAN | Andrew Stanton

1.

a) $2N^2$

b) $4N^2$

c) $2(1-s)N^2$

d) $8(1-s)N^2 + 4(N+1)$

2. $\text{Speedup} = \frac{2N^2}{2(1-s)N^2} = \frac{1}{1-s}$

$2 = \frac{1}{1-s} \Rightarrow \boxed{s = 0.5}$

3. $8(1-s)N^2 + 4(N+1) = 4N^2$

$\boxed{s = 1 - \frac{N^2 - N + 1}{2N^2}}$

4. sparse

$\text{Memory} = (1-0.9)512^2 \cdot 8 + (513) \cdot 4$

$= 211767 \text{ bytes}$

$t_{execution} = 211767 \text{ bytes} / 320 \text{ GB/s}$

$= 6.618 \times 10^{-7} \text{ s}$

dense

$\text{Memory} = 4(512)^2$

$t_{execution} = 1.04858 \times 10^6 \text{ bytes} / 320 \text{ GB/s}$

$= 3.277 \mu s$

$\text{Speedup} = \frac{3.277 \times 10^{-6}}{6.618 \times 10^{-7}} = \boxed{4.95}$